

Project information

Name	Characterizing the complete disk population of Herbig Ae/Be stars within 1 kpc
Code	2024.1.00408.S
PI	Miguel Vioque
Organization	ESO, ESO
Co-Is	A. Booth, S. Grant, M. Hogerheijde, A. Roy, L. Stapper, J. Williams

ObsUnitSet information

Name	Member OUS (IRAS_09245-5228)
QA2 Status	✓Pass
Member OUS Status ID	uid://A001/X3788/Xc194
SchedBlock name	IRAS_092_a_06_7M
SchedBlock UID	uid://A001/X3788/Xc154
Array	7M
Mode	Pipelineable
Band	ALMA_RB_06
Repr.Freq. (sky)	230.54 [GHz]
Spectral setup	ACA
Sources	IRAS_09245-5228
Other SBs in this Group	
OUS (Member OUS	
Status ID in brackets):	
Execution count	2.09 of 1 expected

Final QA2 comment

Data reduction method(s): PipelineCalAndImaging

Calibration issues: None

Imaging issues: None

It is recommended that the PI carefully assess the results on the hif_findcont weblog page, and the "line-free moment 8" images on the cube imaging weblog page.

QA2 was performed on the image whose channel width is the one closest to the representative bandwidth selected by the PI in ALMA OT, in the spectral window closest in frequency to the representative frequency. Both the beam size and the RMS meet the PI requested performance parameters. Therefore, this scheduling block has been deemed a QA2 PASS.

Representative Window -

Image name: uid___A001_X3788_Xc194.s34_0.IRAS_09245-5228_sci.spw28.mfs.regcal.l.iter1.image
 Robust = 0.50
 Beam size = 6.959 x 4.296 arcsec
 RMS = 2.124 mJy/beam over 74.053 MHz
 RMS at PI requested resolution = 0.241 mJy/beam over 5742.188 MHz (7467.53 km/s)

Aggregate Continuum -

Image name: uid___A001_X3788_Xc194.s36_0.IRAS_09245-5228_sci.spw24_26_28_30.cont.regcal.l.iter1.image
 Robust = 0.50
 Beam size = 7.068 x 4.197 arcsec
 RMS = 0.291 mJy/beam over 4.711 GHz

For additional information on the calibration and imaging pipeline products please see the Knowledgebase article: <https://help.almascience.org/kb/articles/what-calibration-and-imaging-products-will-be-delivered-to-me>

RMS and beam size at representative frequency

Sensitivity goal	0.350 [mJy] over bandwidth 5.742 [GHz]		
Angular resolution goal	N/A		
Achieved RMS			
for desired bandwidth	0.241 [mJy]	for continuum	N/A
Synthesized beam	Mean (arcsec)	5.468	
Major axis (arcsec)	6.959	Minor axis (arcsec)	4.296
		Position angle (deg)	-81.154

Execution blocks summary													
EB	N Ant.	Start Time	End Time	ToS (sec)	Avg. Elev. (deg)	Trans. Elev.	Mean PWV (mm)	Phase RMS (deg)*	Min BL (m)	Max BL (m)	AR (")	MRS (")	EF
uid://A002/X11e3e46/Xff4a	10	2024-10-02 13:09:	2024-10-02 13:58:	2974	59.9	60.3	0.0	16.648	8.9	48.9	4.5	24.6	0.59
uid://A002/X11eaef9/X7c3a	12	2024-10-12 14:22:	2024-10-12 15:10:	2878	50.5	60.3	0.4	12.022	8.9	84.3	3.1	20.1	1.50

*WVR-corrected value averaged over all basebands and scans.

Pipeline executions			
CASA version	Pipeline version		Report date
6.6.1.17	2024.1.0.8		2024-10-29 18:20:15
			2:35:28

Calibrator source fluxes				
Source name	Flux (Jy)	Flux error (Jy)	Frequency (GHz)	Intents
J0904-5735	1.148	0.115	216.491	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.148	0.115	219.491	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.136	0.114	230.529	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.128	0.113	233.991	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.059	0.106	216.492	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.056	0.106	219.492	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.034	0.103	230.531	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0904-5735	1.036	0.104	233.992	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR

Spectral Windows			
Transition	Central Frequency (sky, GHz)	Bandwidth (GHz)	N of channels
LSB (SW-1)	216.488	1.875	1920
USB_COiso (SW-1)	219.488	1.875	1920
USB_CO (SW-1)	230.527	0.117	1920
USB (SW-1)	233.988	1.875	1920

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INTRODUCTION

ALMA currently provides data products for each member observation unit set (MOUS), corresponding to one or more executions of a single scheduling block (SB). In all cases except for polarization projects, each execution is independently calibrated, and then all executions are imaged together.

The following text describes the contents of this data package. Further details and updates can be found in the ALMA Knowledge Base (KB) articles indicated throughout this document.

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DATA DELIVERY CONTENTS

Each data package tarball expands to contain a single member.<ous_id> directory with the following sub-directories: calibration, script, qa, log, product. (Note: The ALMA archive will have prepended many product and script names with the MOUS UID, compared to the names displayed in the pipeline's weblog.)

- 'qa/' contains a number of files that summarize the quality of the observations and products. There is a "QA0 report" for each execution giving an overview of the conditions (Tsys, phase RMS) and quality of each observation. The "QA2 report" (this file) provides an overall assessment of the complete observation and final products. For pipeline-processed data there is a member.<ous_id>*.weblog.tgz file for each time the pipeline was run (calibration, imaging, or calibration+imaging) that unpacks into "weblog" (open the pipeline-<DATE>/html/index.html file in a web browser), which is a series of webpages that present the details of the pipeline run, many diagnostic plots, and quality metrics. For manually calibrated products, several text files and png images are provided instead, which can be used in conjunction with the *scriptForCalibration.py described below to understand how the data were calibrated.

- 'product/' contains fits files of the imaging products. In some cases, only a subset of all science targets/spectral windows are imaged, to demonstrate the quality of the calibration and images. If the products were created by the imaging pipeline, a detailed comprehensive explanation of what images were created, how continuum was subtracted, and the quality of the processing is contained in the pipeline weblog (see above). If images were created manually, please refer to the *scriptForImaging.py described below for details of the process.

- 'script/' contains the scripts to:

- calibrate raw visibilities (*casa_pipescript.py in the case of pipeline calibration, and *scriptForCalibration.py in the case of manual calibration).

- restore calibrated visibilities without fully rerunning the pipeline (*casa_piperestorescript.py)

- restore calibrated visibilities ("*scriptForPI.py"; works for either pipeline or manually calibrated data - see the section "How to restore the calibrated Measurement Set (MS) for your data" further below).

- create images (*casa_pipescript.py in the case of pipeline imaging, and *scriptForImagingPrep.py and *scriptForImaging.py in the case of manual imaging).

- In case the calibration was done by the automated pipeline, you will also see the Pipeline Processing Request file (PPR).

(Note: the ALMA archive may prepend the script names with the MOUS UID.)

- 'calibration/' contains the files needed for calibration, including calibration and flagging tables and manually introduced flagging commands, which allow the user to start from the initial ASDM files and obtain the fully calibrated data.

- 'log' contains the CASA log files.

For more information see the ALMA QA2 Data Products document (<https://almascience.org/processing/qa2-data-products>), the ALMA Technical Handbook (in particular chapter 11 on the QA2 pass criteria), and the relevant Call for Proposals for this Cycle, which are available for download from the ALMA Science Portal at <https://almascience.org/documents-and-tools>.

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INTERFEROMETRIC CALIBRATION AND IMAGING REGENERATION

This section concerns ALMA interferometric (12m Array and ACA) data. The ALMA KB article "How to Restore the Calibrated Measurement Set (MS) for your Data" at <https://help.almascience.org/kb/articles/how-do-i-obtain-a-file-of-calibrated-visibility-measurement-set-for-alma-data> will contain an up-to-date version of the procedures described below.

How to Restore the Calibrated Measurement Set (MS) for your Data

Calibration is the first step of processing interferometric data. Many users will want to recreate the calibrated measurement set and then perform the second step (imaging) using their own scripts, or by modifying the imaging script provided with the data.

In order to obtain your calibrated data, download the raw data in ASDM format from the science archive. If you downloaded and untarred all available files for this delivery as described in the notification email, then you will already see these files (with names like "uid*.asdm.sdm") in an additional sub- directory called "raw". Otherwise, you will need to download and untar the tar balls of the raw data in the same directory that you untarred the tarball of the products, creating the "raw" sub-directory and associated ASDM files.

For the next step, you will need the right version of CASA to be installed. Please find the line starting with "CASA version used for reduction:" in your QA2 report or README. The version indicated there is what you need to use for running the scriptForPI.

Once the raw data is in place, cd to the "script" sub-directory, start CASA:

```
casa --pipeline
```

(Note: For some versions, CASA is released without pipeline and the "--pipeline" switch is not available. You will need a version with the pipeline if there is a file named "PPR*.xml" or "*pprequest.xml" in your script/ sub-directory. If there is no such file, you will not be able to run the the pipeline or restore pipeline-calibrated data. For more information on the execution of the pipeline, please refer to the ALMA Science Pipeline User's Guide available at <https://almascience.org/documents-and-tools/pipeline-documentation-archive>).

Then look for the script ending in "scriptForPI.py" and execute it, e.g.:

```
execfile("member.uid___A001_X123a_X45.scriptForPI.py")
```

Running the scriptForPI will result in one or more calibrated measurement sets (MSs) ready for imaging. You can use the uid_*.ms directly, keeping in mind that the calibrated data is stored in the CORRECTED data column, and all sources (including calibrators) are in the MS. If you want to run the imaging pipeline, you should run it from this point, without changing the uid_*.ms.

Imaging

If the data were imaged with the pipeline, many details of what images were created, how and why, are contained in the pipeline weblog. Each "task" of the pipeline weblog corresponds to a pipeline "hif_" or "hifa_" task which the user can run manually to reproduce what the pipeline did, optionally changing parameters. The full sequence of such tasks run by the pipeline are contained in "script/*casa_pipescript.py", which could be run to completely reproduce all pipeline processing. Alternatively, the calibrated measurement set produced above can be used as the basis for running only the imaging tasks in the pipeline. See https://casaguides.nrao.edu/index.php/ALMA_Imaging_Pipeline_Reprocessing for details.

Each imaging section of the weblog also links to the CASA commands that were used by the pipeline – these form a good initial template which users can modify to manually image the data with different parameters.

If the data were manually imaged, refer to "script/*scriptForImaging.py", which contains the CASA commands that were used to create the image products from the calibrated MS. The "*scriptForImaging.py" may partially be interactive (for masking) and should be executed by copy and paste.

Options

1) If you want to force scriptForPI.py to split out the science spectral windows, you need to set the variable DOSPLIT=True before starting the scriptForPI in CASA.

2) The scriptForPI will usually run *casa_piperestorescript.py, which applies the calibration and flagging tables from the calibration/ sub-directory to the raw MS, restoring a calibrated MS. If *casa_piperestorescript.py is not available (as is the case for manually-calibrated datasets), the scriptForPI will instead run the entire calibration script, re-creating the calibration and flagging tables. You can force the execution of *casa_pipescript.py to rerun the entire calibration pipeline (instead of only restoring the existing calibration) by moving *casa_piperestorescript.py out of the script directory. Rerunning the calibration can be useful if you want to tweak its parameters. Otherwise the restore is faster.

3) The scriptForPI offers some global variables for your convenience. You can find an explanation in this Knowledgebase Article: <https://help.almascience.org/kb/articles/what-global-variables-are-available-in-scriptforpi-py>

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PRIMARY BEAM CORRECTION

The images included in delivery are corrected for the primary beam (PB) response, i.e. the dependence of the instrument's sensitivity on direction within the field of view.

For each image, two files are being delivered:

- a) the PB-corrected image (file name ending in ".pbcor.fits")
- b) the image of the PB which was used in the correction (ending in ".pb.fits" or ".flux.fits", typically gzipped in which case you need to unzip the file before using it)

The image noise was measured in the uncorrected image. The corrected image (a) was then obtained by dividing the uncorrected image by the PB image (b). The uncorrected image can be recovered using the CASA task impbcor in mode "m":

```
impbcor(image='image.pbcor.fits', pbimage='image.pb.fits', mode='m', outfile='image.recovered')
```

See also the ALMA KB article "Where is the Primary Beam Correction Information in my Delivered Data" at

<https://help.almascience.org/kb/articles/where-is-the-primary-beam-correction-information-in-my-delivered-data> for any updates to this procedure.

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